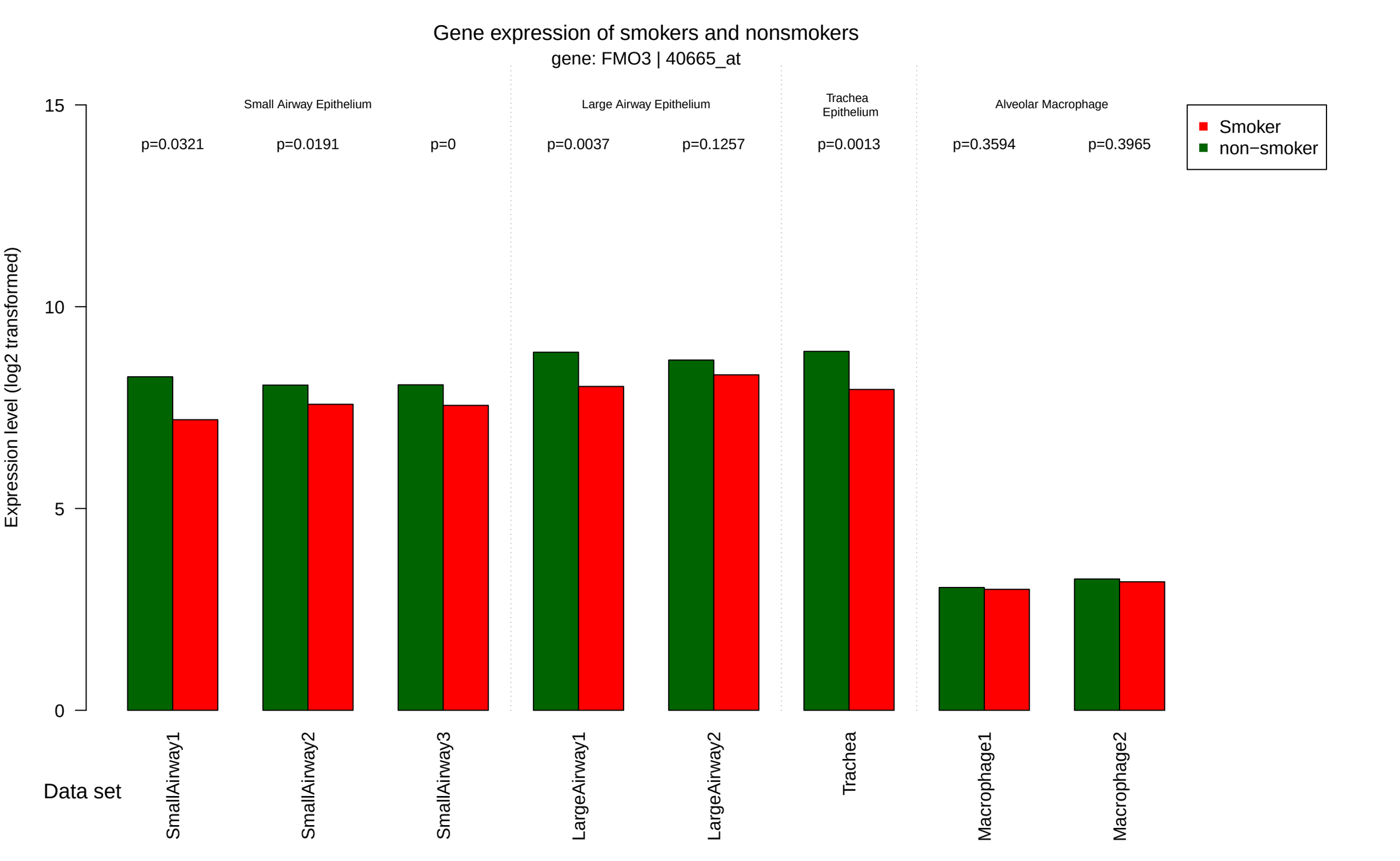
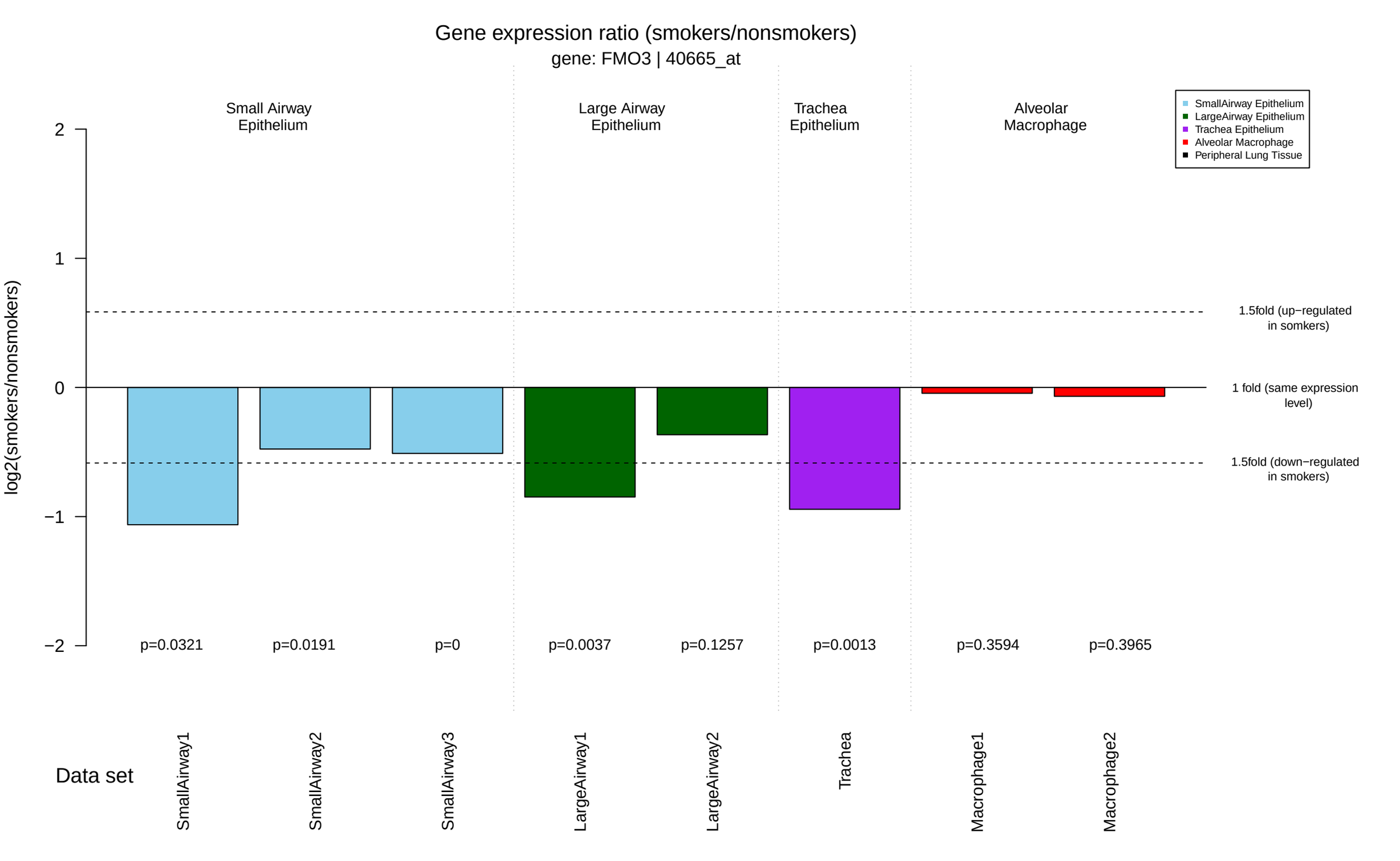
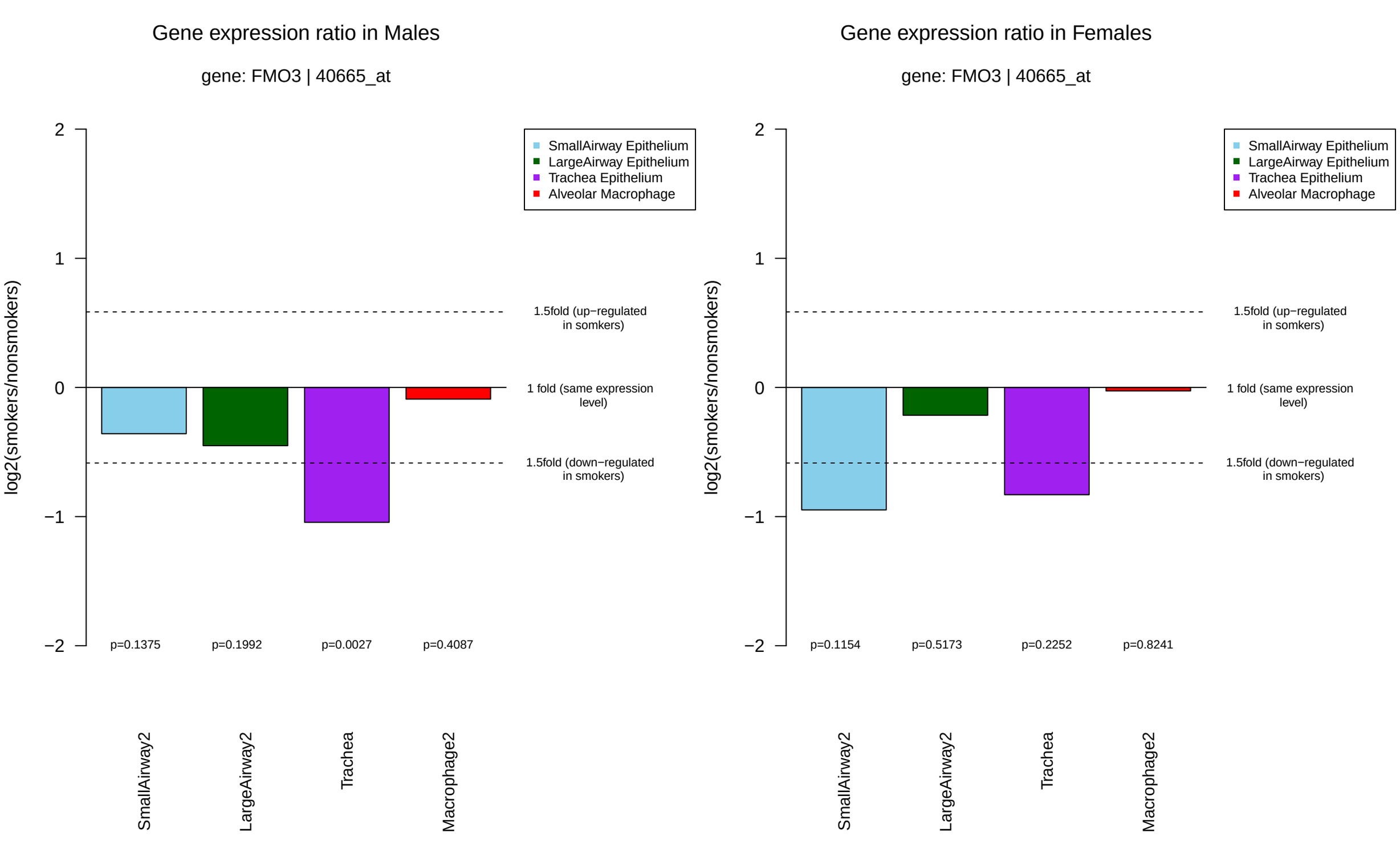
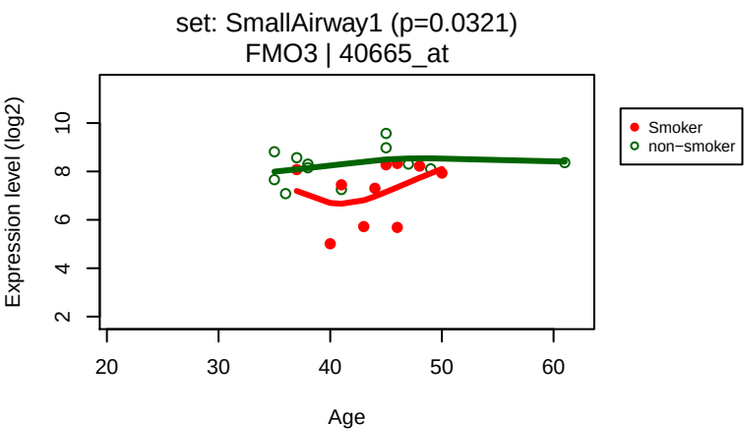
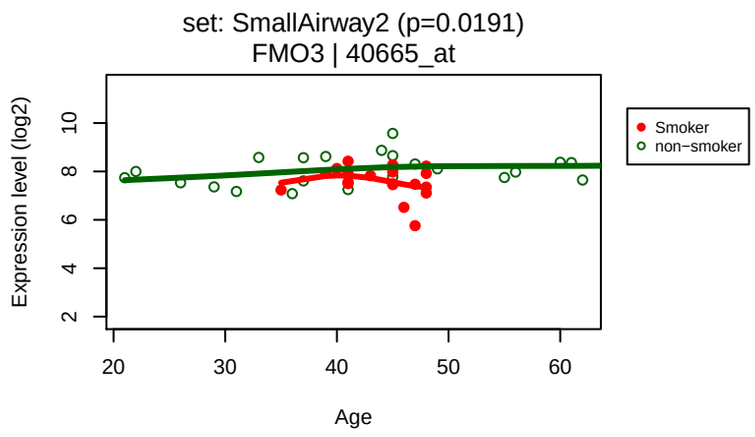
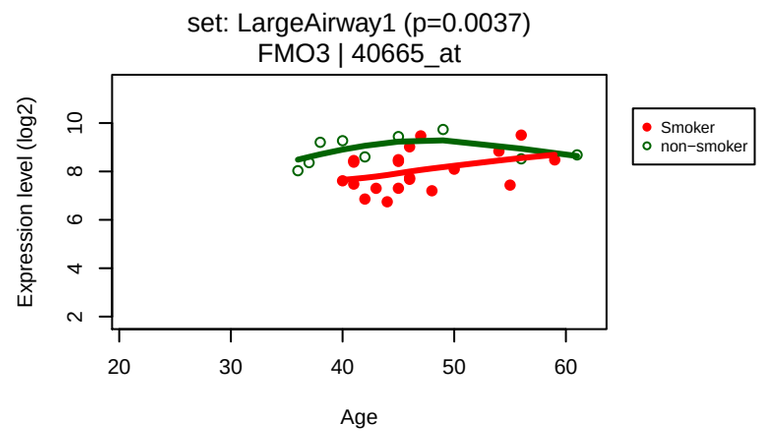

Gene expression of smokers and nonsmokers
gene: FMO3 | 40665_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0321
p=0.0191
p=0
p=0.0037
p=0.1257
p=0.0013
p=0.3594
p=0.3965
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

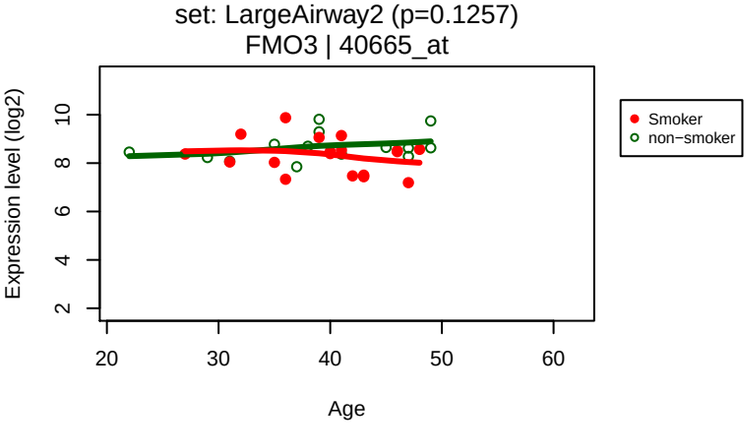

Gene expression ratio (smokers/nonsmokers)
gene: FMO3 | 40665_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0321
p=0.0191
p=0
p=0.0037
p=0.1257
p=0.0013
p=0.3594
p=0.3965
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

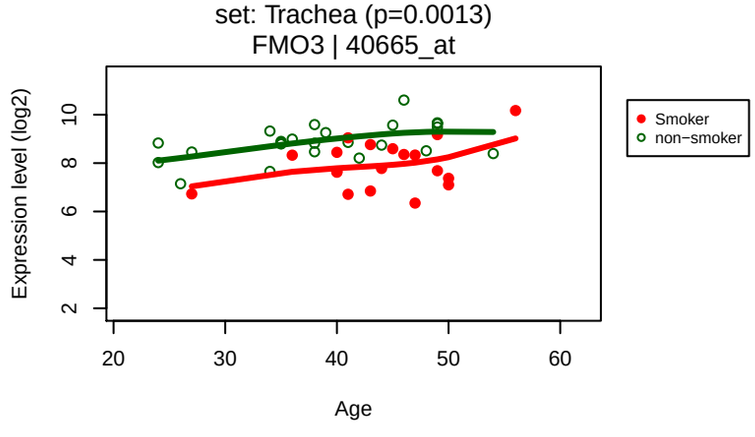

Gene expression ratio in Males
gene: FMO3 | 40665_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1375
p=0.1992
p=0.0027
p=0.4087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FMO3 | 40665_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1154
p=0.5173
p=0.2252
p=0.8241
SmallAirway2
LargeAirway2
Trachea
Macrophage2

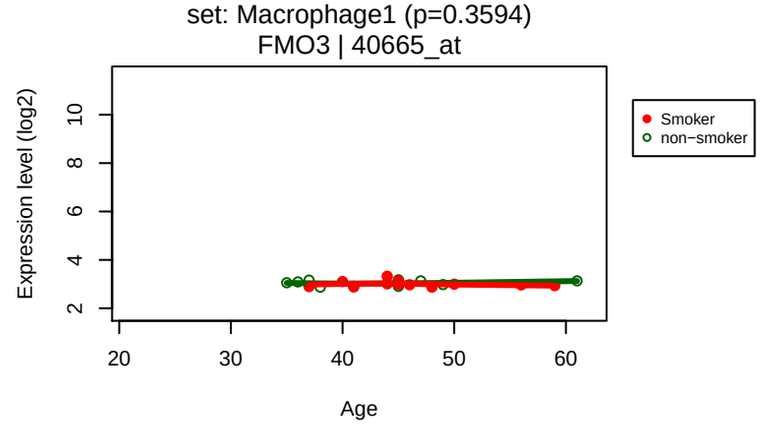

set: SmallAirway1 (p=0.0321)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker

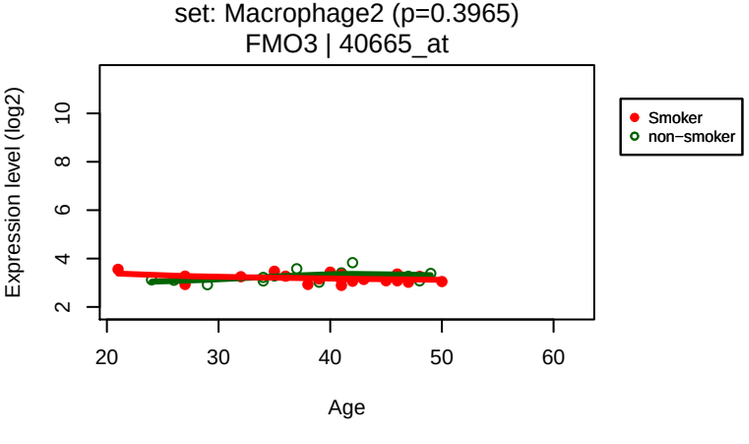

set: SmallAirway2 (p=0.0191)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker

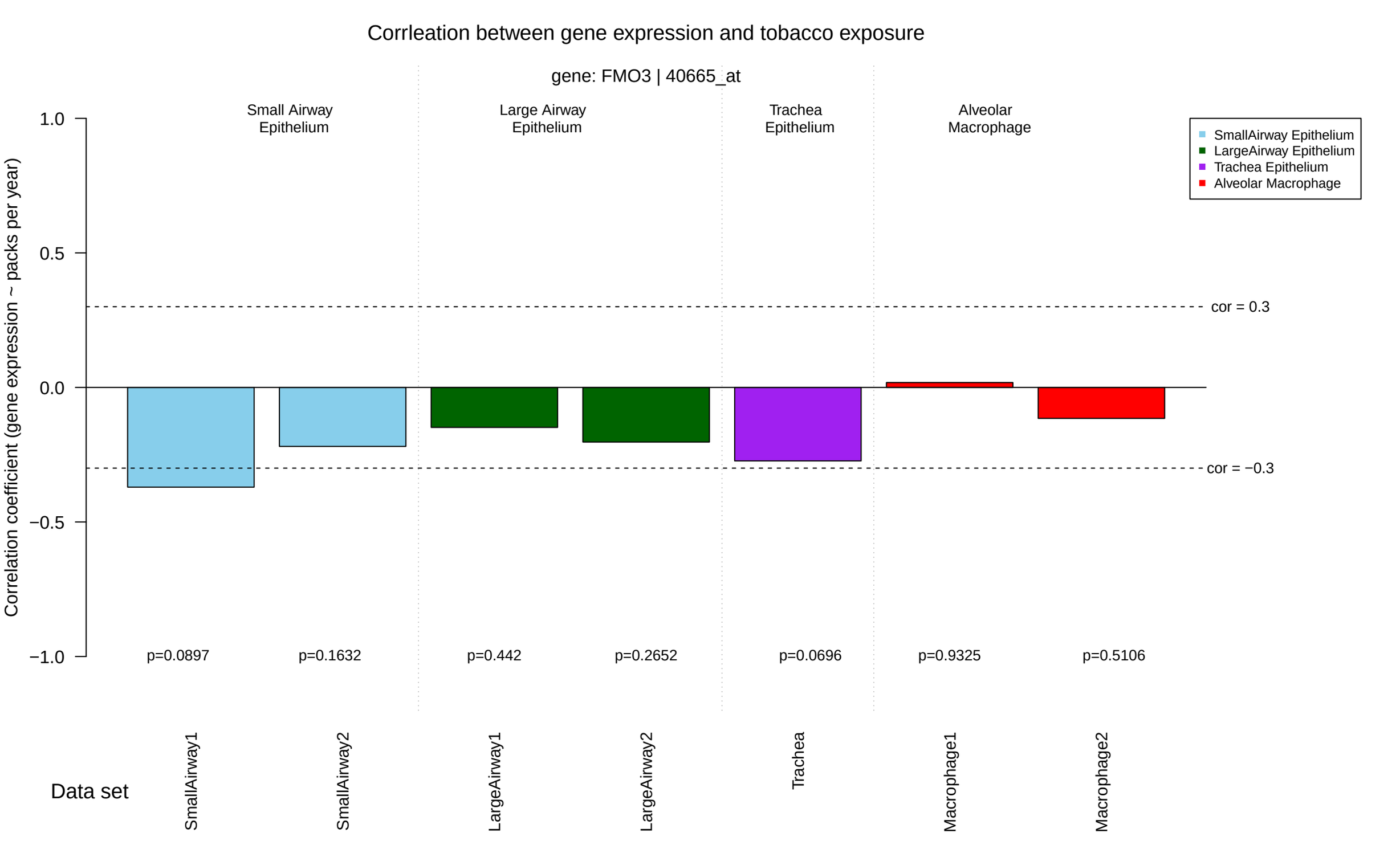

set: LargeAirway1 (p=0.0037)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1257)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3594)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3965)
FMO3 | 40665_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FMO3 | 40665_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0897
p=0.1632
p=0.442
p=0.2652
p=0.0696
p=0.9325
p=0.5106
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2